BrandCheck

Pre-publication brand governance: multimodal content review, source-grounded AI, and accountable human decisions.

ONE-MINUTE JUDGE JOURNEY (NO SETUP REQUIRED)

1. Explore instantly: Open /example for a zero-login interactive demo. 2. Multi-tenant workspace: Sign in at /workspace to test real multi-user roles. 3. Document review: Import PDF/DOCX at /documents with page/paragraph navigation. 4. Human handoff: Creator assigns a reviewer; creator self-approval is denied by the server.

HOSTED WORKING PRODUCT

https://brandcheck-codestar-judge.freelynx19.chatgpt.site/workspace

PUBLIC JUDGE SOURCE BRANCH

https://github.com/Manas-Jain-24/brandcheck/tree/judge-hosted

DEPLOYED PRODUCT INTERFACE: SHARED REVIEW & VISUAL LEADS

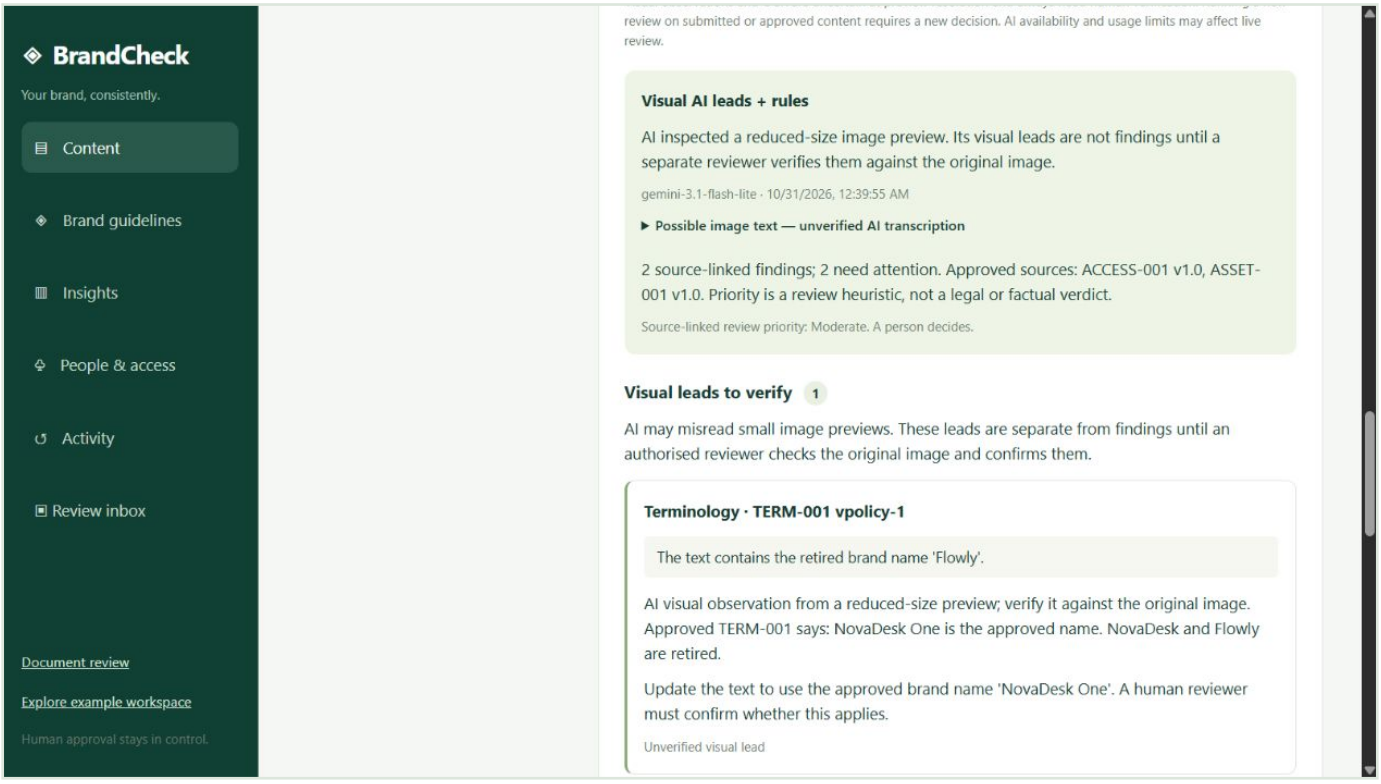


Figure 1: Deployed multi-tenant workspace showing deterministic findings, grounded Gemini review, and unverified visual leads awaiting reviewer confirmation.

01 / DELIVERED ROUND 3 IMPROVEMENTS

Execution of Promised Capabilities

In Round 2, our team committed to three major upgrades: shared authenticated governance, document and visual review, and policy connectors with regression testing. In Round 3, we delivered working implementations for all three, engineered with strict transactional boundaries.

IMPROVEMENT 1: AUTHENTICATED SHARED WORKSPACES & GOVERNANCE

Delivered: Multi-tenant Cloudflare D1 (SQLite) backend with platform-authenticated sessions. Server-enforced roles (Admin, Creator, Reviewer) and persistent drafts. Anti-self-review enforcement: Creators cannot approve or decide their own submissions (HTTP 403). Optimistic concurrency control (revision checks) and append-only database audit logs for all events and snapshots.

IMPROVEMENT 2: MULTIMODAL DOCUMENT PARSING & VISUAL GOVERNANCE

Delivered: Client-side PDF (PDF.js) and DOCX (Mammoth/fflate) text extraction with location-linked findings mapped directly to PDF pages and Word paragraphs. Editable section text, safe zip decompression protection against zip bombs, and seamless workspace handoff. Visual governance checks exact asset SHA-256 fingerprints, alt-text presence, contrast ratios, and dominant palette heuristics. Opt-in Gemini vision review surfaces unverified visual leads that do not affect scores until an

IMPROVEMENT 3: GITHUB POLICY CONNECTOR & ATOMIC REGRESSION ENGINE

Delivered: Real on-demand read-only GitHub raw JSON connector for brand guidelines. Structured proposal lifecycle with visual diffing against active policy. Atomic regression rechecking: Activating a policy executes a 5-statement atomic transaction using SQLite's json_each, evaluating all stored workspace drafts simultaneously. Content affected by rule changes is automatically transitioned back to needs_review status with full before/after diff audit history.
Deliberate MVP Scope: Public GitHub JSON connector (Notion/Confluence/Google Docs integrations and NLP rule extraction

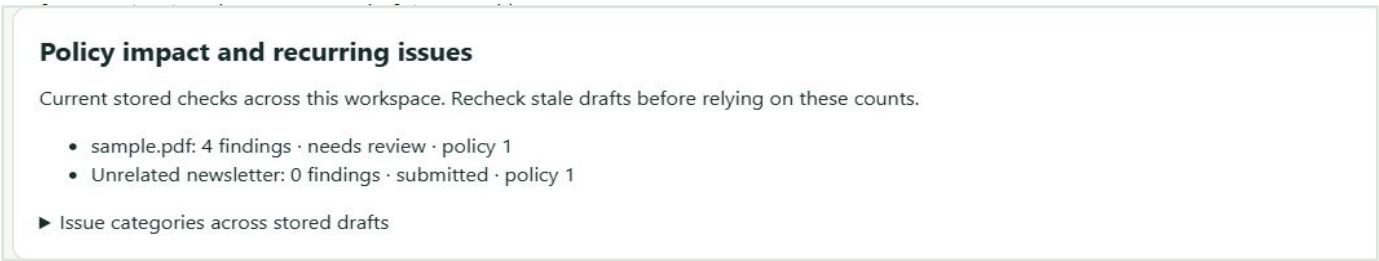


Figure 2: Verified production policy regression: activating a renamed term rechecked drafts in an atomic transaction; affected drafts automatically reverted to 'needs review'.

02 / ENGINEERING ARCHITECTURE & PROOF

Transactional Integrity & Verified Tests

Rather than treating AI as a black box, BrandCheck pairs serverless edge performance with strict schema validation, optimistic concurrency control, and deterministic rule fallbacks.

FULL-STACK ARCHITECTURE & DATA FLOW

Edge Runtime & UI: React 18, TypeScript, Vite, Tailwind CSS running on Cloudflare Workers.
Persistence & Schema: Cloudflare D1 (serverless SQLite) with additive Drizzle ORM migrations. Zero runtime table creation.
Grounded AI Pipeline: Browser $\rightarrow$ Same-Origin API $\rightarrow$ Private Runtime Secret $\rightarrow$ Google Gemini 3.1. Draft excerpts must physically exist in submitted text and map to an active approved source. Unmapped model claims are filtered out.
Safe Ingestion: Client-side PDF.js worker; Mammoth with fflate zip-expansion limits; 160 KB API request streaming guard.

CONCURRENCY & ANTI-SELF-REVIEW ENGINEERING

- Atomic Compare-and-Swap (CAS): Draft updates and review decisions enforce exact revision matching (version = version + 1 WHERE version = ?). Concurrent mutations return HTTP 409 Conflict without data corruption.
- Anti-Self-Review Guard: The server verifies draft.author_id !== user.userId on every review decision and visual lead confirmation. Creators cannot rubber-stamp their own content via UI or direct API manipulation.
- Graceful AI Fallback & Quotas: High-volume quota allowance (180 requests/day, 20/10-min IP bucket). When AI is unreachable or exhausted, deterministic term/alt-text/claim rules execute seamlessly with clear provenance labeling.

VERIFICATION RECORD (46 AUTOMATED TESTS PASSING)

- SQLite Service Suite: 46 comprehensive service/helper tests pass locally using actual D1 migrations and in-memory SQLite.
- Tested Scenarios: Workspace isolation, role boundaries, self-review denial, reviewer assignment, CAS concurrent writes, transaction rollback, policy regression, document location mapping, and visual lead confirmation.
- Real Document Fixtures: Production validation tested with actual 2-page PDF and zipped XML DOCX files.
- Strict Type Safety: Strict TypeScript build passed with zero errors (tsc --noEmit).

GOVERNANCE TOOLING & ENTERPRISE FEATURES

Includes an exportable Brand Clearance Certificate, a live Pre-Flight Status Matrix, 1-click Batch Terminology Corrections, Reviewer Decision Templates, and a smart WCAG 2.2 Contrast Optimizer with auto-adjusting compliant color calculation.

Boundaries, Disclosures & Contributions

We believe in absolute intellectual honesty. The capabilities described in this report represent verified, functional software. Features outside our MVP scope are explicitly declared below.

WHAT DOES NOT WORK YET (DELIBERATE MVP BOUNDARIES)

- Document Extraction: PDF and DOCX parsing requires extractable text. Scanned images have no OCR; documents are limited to 15 MB, 40 PDF pages, and 80,000 extracted characters. Workspace drafts support up to 10,000 characters.
- Visual Governance: Exact asset SHA-256 fingerprint matching, alt-text presence, text contrast ratio, and dominant palette heuristics are supported. Full-image logo clearance distances, trademark placement, and automated canvas coordinate boxes are not implemented.
- Policy Synchronization: Supports on-demand public GitHub raw JSON import with human approval. Automated background polling, private OAuth connectors (Notion/Confluence/Google Docs), and NLP memo interpretation remain future work.
- Audit Ledger: Audit events are stored in Cloudflare D1 with trusted administrators; this is an application append-only log, not a cryptographically signed blockchain or SOC2 certification.

THIRD-PARTY REUSE & AI ASSISTANCE DISCLOSURES

Reused Components: The foundational web structure utilizes the Sites Vinext/Vite starter on Cloudflare Workers. Document parsing incorporates PDF.js (Apache-2.0), Mammoth (BSD-2-Clause), and fflate (MIT).
AI Models & Coding Assistance: Multimodal semantic analysis calls Google Gemini 3.1 Flash Lite with private server credentials. Significant AI coding assistance was utilized during development for rapid prototyping, refactoring, and test fixture authoring.

VERIFIED TEAM MEMBER CONTRIBUTIONS

Manas Jain (Team Leader & System Architect):
Responsible for overall system architecture, Cloudflare D1 database design and additive Drizzle migrations. Implemented the atomic compare-and-swap (CAS) mutation engine, server-enforced role permissions, and anti-self-review guards. Designed the SQLite json_each policy regression recheck engine, integrated Google Gemini with excerpt grounding, built the primary React/TypeScript UI, and managed production deployment.

Mayank Kumar (Teammate & QA / Testing Specialist):
Responsible for end-to-end test suite design, test fixture authoring (synthetic 2-page PDF and XML-based DOCX files), and boundary condition validation across the 46 SQLite-backed service tests. Validated the public GitHub policy connector, conducted cross-account browser QA verifying separate creator and reviewer sessions, authored testing documentation, and ensured compliance with CodeStar 30 submission guidelines.

FINAL SUBMISSION STATEMENT

First product commit: 26 October 2026, 1:49 AM IST. All custom code, schemas, and UI components were developed specifically for CodeStar 30. No proprietary Inkloom AI codebase was used. Deployed live on Cloudflare.